

UMSL ME NTT Interview Questions and Responses

Q1: Why did you choose UMSL?

There are really two layers to my answer, professional and personal, and they reinforce each other. On the professional side, I'm already embedded in the UM system. I've been at Missouri S and T, so I understand the culture, the administrative structures, and the expectations of a UM institution. I also have working relationships with colleagues at UMKC and other campuses through my STEM education research, so UMSL wouldn't feel like starting from scratch. It would feel like moving deeper into a network I already belong to. Beyond the system itself, UMSL specifically appeals to me because of who your students are. This is an urban, commuter-heavy, first-generation-heavy institution, and my research on course prerequisite networks showed me that these are exactly the students most affected when curricula aren't designed with progression and attrition in mind. I want to teach where that work actually matters at the course delivery level. I'm also genuinely excited about the direction of the School of Engineering. The Design and Innovation Studio opening in Fall 2026, the CNC equipment, the wind tunnel, the MTS testing machine, that's the kind of infrastructure that makes experiential engineering education real, not aspirational. Being part of building curriculum around that facility from the ground up is the kind of opportunity that doesn't come around often. On the personal side, my family is here. My wife is a faculty member in the region, my children are in school here, and our support network is rooted in St. Louis. This isn't just a convenient coincidence. It means I'm genuinely invested in this community long-term, and UMSL is where I want to build my teaching career.

Q2: Tell us about your previous teaching experience.

I'll give you the honest arc rather than just a list of courses. I started teaching in Spring 2019 as a TA at Missouri S and T, and my first semester was humbling. I was competent in the material but I hadn't yet learned the difference between knowing something and teaching it. By the time I finished, I had taught 35 sections across six years, Fluid Mechanics, Heat Transfer, Internal Combustion Engines, Aerospace Experimental Methods, and Linear Algebra as a sole instructor at the 3000-level. Over 900 students total. My effectiveness scores started at 3.0 out of 4.0 and trended upward into the high 3s, which I'm proud of because it reflects real course-by-course adjustment, not luck. What I think is most relevant for UMSL specifically is that I've taught both ends of the difficulty spectrum you're dealing with. On one end, gateway-level fluid and thermo, courses where foundational gaps sink students before they reach the interesting material. On the other end, upper-division lab courses where students have to integrate theory, instrumentation, and judgment simultaneously. Both require completely different pedagogical approaches, and I've had to develop both. Looking at your curriculum specifically, I can cover the full Fluid and Heat Transfer sequence from ENGR 2330 through MENG 3370, 3380, and up to MENG 4450 in CFD. That's a complete vertical. I can also teach across the Solid Mechanics, Manufacturing, and Engineering Core tracks. The two areas I haven't formally taught yet, composites and the controls lab, I can prepare for within a semester, and I'm honest about that distinction.

Q3: How do you do student retention and recruitment?

I think about these as two distinct problems that require different strategies, though they share a common foundation. Students stay and students come when they feel like they belong and can succeed. On the retention side, my research gives me an advantage most instructors don't have. My published work analyzing prerequisite networks across ME and CS programs directly maps which gateway courses carry the highest attrition risk. I've seen the data. Students who struggle in foundational math and mechanics courses face compounding disadvantages throughout the degree. So in my own courses, I treat those foundational connections as instructional priorities. I identify at-risk students early through formative assessment, hold structured office hours, and build in explicit prerequisite bridges rather than assuming prior knowledge held. The DEEP project, which I've studied as part of my curriculum research background, made a related point. Prerequisite chain length is itself a retention barrier, particularly for students who enter programs later or transfer in. That framing directly informs how I advise students. If a sophomore comes to me wondering whether it's too late to pursue ME, the honest answer has to be grounded in knowing where the real time constraints are in the curriculum. On the recruitment side, I've done direct outreach work. I volunteered in a robotics workshop in East St. Louis, working with Upward Bound high school students who wouldn't typically see themselves as future engineers. I also taught 3D Modeling and 3D Printing courses through SIUE's community education program in 2018 and 2019, reaching adult learners outside the traditional pipeline. Going forward, I'd want to build on those connections. Formalize high school pipeline partnerships, engage with FIRST Robotics teams, and work with the department on community college transfer advising. UMSL's location in St. Louis makes that kind of outreach genuinely accessible.

Q4: Tell us about your previous lab development experience.

My lab experience has two distinct layers, managing existing labs and developing new lab content, and they've informed each other. At Missouri S and T I managed the Internal Combustion Engines lab and the Aerospace Experimental Methods lab. That meant not just running experiments but maintaining documentation, aligning lab procedures to learning outcomes, and troubleshooting equipment mid-semester when things inevitably break. You learn very quickly what a lab is actually testing versus what you think it's testing. At SIUE I ran the Fluid Mechanics and Heat Transfer labs while simultaneously conducting PIV research in the same facility. That dual role, researcher and lab instructor in the same space, was unusual and valuable. I could connect what students were doing in lab directly to what active research looks like, and that changes the atmosphere of the room considerably. In terms of development work, I've contributed to rubric design, outcome alignment documentation, and the kind of pre-lab and post-lab structure that makes experiments genuinely instructional rather than just procedural. My quality assurance background from aviation, ISO 9001 processes, systematic documentation, root cause analysis, has given me a framework for thinking about lab procedures as systems that can be improved, not just followed. I'm also aware of the Design and Innovation Studio opening in Fall 2026, the VF-2 CNC, the wind tunnel, the MTS universal testing machine, the 3D printing stations. My background in fluid mechanics, manufacturing, and materials testing positions me well to help develop lab curriculum for that space. Being part of building that curriculum from the ground up is something I'd actively want to contribute to.

Q5: How do you mentor students outside of class?

My approach to mentoring has evolved from being reactive, answering questions when students showed up, to being genuinely proactive, and that shift matters more than people acknowledge. The most consistent thing I do is learn students' names and goals early. Student feedback has specifically noted it. One student wrote, only teacher to remember my name. When you know what a student is working toward, you can connect what's happening in class to their actual trajectory, and that's what makes mentoring real rather than transactional. Outside of class, I work with students at a few different levels. For students who are struggling, I don't just rework problems. I try to find where in the prerequisite chain their understanding broke down. A student failing thermodynamics is often actually struggling with something from calculus or physics that was never properly resolved. Identifying that is worth more than re-explaining the same thermo concept a third time. For stronger students, I've helped with graduate school navigation, letter-of-recommendation coaching, and making introductions to research faculty and industry contacts. One of my goals for any student who works with me closely is that they leave with a clearer picture of what their options are, not just a grade. I'm also applying to serve as a mentor in the LSU Virtual Math Research Circle, which pairs mentors with small teams of high school students for structured six-week research projects with a real deliverable. That's a model I find compelling, mentorship with structure and accountability. At UMSL specifically, with your FIRE-E and SURE programs, I see natural structures I'd want to plug into from day one.

Q6: Tell us about your student club supervision activities.

I'll be transparent. My direct club supervision experience is limited. I haven't formally advised a student organization yet. What I have done is engaged with students in adjacent spaces: outreach events, robotics workshops, design activities outside the formal curriculum. What I can speak to more honestly is what I would bring to club supervision and what I'd want to build. In ME programs, the most high-impact student clubs tend to be ASME student chapters, SAE teams, and design competition teams. These organizations work best when faculty advisors are genuinely engaged rather than just signing off on paperwork. My background, spanning fluid mechanics, manufacturing, materials, and CAD, means I could be a substantive technical resource for a senior design competition team or a formula-style SAE group. What excites me specifically about UMSL is the Design and Innovation Studio opening in Fall 2026. A facility with CNC equipment, a wind tunnel, 3D printing, and a universal testing machine is exactly the kind of infrastructure that makes a student design team competitive and meaningful. If UMSL has an active ASME chapter or is looking to build one out, anchoring it around that facility, with connections to your FIRE-E and SURE undergraduate research programs, would create something with real depth. I want to be clear that this would be a learning curve for me in the administrative sense, and I'd want to connect with experienced club advisors in the department to do it right. But the technical engagement and student investment are areas where I can contribute immediately.

Q7: What is your ABET assessment experience?

I came to ABET from two directions, practical lab management and curriculum research, and together they give me a fairly complete picture. On the practical side, I've managed laboratory courses at both Missouri S and T and SIUE, Internal Combustion Engines, Aerospace Experimental Methods, Fluid Mechanics, and Heat Transfer. In all of those, I was working directly with outcome-based

assessment: writing rubrics tied to specific student outcomes, collecting and documenting student artifacts, and feeding results into the continuous improvement cycle. I understand that ABET accreditation isn't a documentation exercise. It's a quality improvement system, and closing the loop is what makes it meaningful. On the research side, my prerequisite network work required me to engage deeply with ABET-accredited programs across Missouri S and T, SIUE, SIUC, UIUC, and UMKC. Studying how those programs structured their curricula to satisfy the seven EAC student outcomes gave me a systems-level view of how course-level decisions aggregate to program-level accreditation health. I'm also familiar with the upcoming 2026 ABET criteria changes addressing AI literacy and data-driven engineering competencies. Those changes ask programs to demonstrate that students can work with computational tools and datasets, and that's a curriculum design challenge I'd want to help address proactively rather than reactively. My quality assurance background from aviation, ISO 9001 certification, auditing, root cause analysis, provides a complementary framework. The language is different but the logic is identical: systematic, evidence-based cycles of assessment and improvement. I'm prepared to attend ABET workshops and serve on assessment committees from day one.

Q8: What is the most important thing you have learned being an ME instructor?

The most important thing I've learned is that the gap between knowing something and teaching it is much wider than I ever expected, and that closing that gap is a fundamentally different skill than mastering the content itself. I came into teaching with strong technical preparation. I knew thermodynamics, fluid mechanics, the mathematics. What I didn't know was how to track where a student's understanding had diverged from mine, or how to find the specific point in their reasoning where the logic broke down rather than just showing them the right answer again. Learning to do that well took years and some genuine failures. The second thing, and this still surprises me, is how much the instructor's presence outside of lecture matters. Not office hours in the transactional sense, but whether students believe you actually care whether they succeed. The feedback that has meant the most to me wasn't about my explanations or my examples. It was: one of the few instructors that has shown himself to care about his students not only in regards to the course but in life as well. That's not something you can fake, and it's not something you can manufacture with policy. What this means for UMSL specifically is that I'm not coming in with a fixed model of what good teaching looks like. I have a track record, I have data, and I have principles, but I'm still learning. The best teaching faculty I've observed are the ones who stay in that posture rather than treating their first few years of experience as the finished product.